Section 35: QoS

Layer: Multi | Reference: Odom Vol2 Ch25: Introduction to QoS | *Original study notes — CCNA 200-301*

Quality of Service (QoS) prioritizes certain types of traffic over others when network resources are constrained, ensuring latency-sensitive applications like voice and video aren't degraded by less time-sensitive bulk traffic.

Key Concepts

- QoS only matters when there is congestion/contention for a limited resource (bandwidth, buffer space) - on an uncongested link, QoS configuration has no visible effect
- Classification identifies which traffic belongs to which category (e.g., voice, video, bulk data), typically using DSCP, CoS, or access-list-based matching
- Marking sets a field in the packet/frame header so downstream devices can quickly identify the traffic category without re-classifying it themselves
- DSCP (Differentiated Services Code Point) is a 6-bit field in the IP header used for Layer 3 marking; CoS (Class of Service) is a 3-bit field in the 802.1Q tag used for Layer 2 marking
- Common DSCP/CoS application mappings: voice traffic typically uses EF (Expedited Forwarding); video commonly uses AF4x classes; best-effort data uses DSCP 0 (Default)
- Queuing determines the order packets are transmitted from an interface when there's a backlog; priority queuing/LLQ (Low Latency Queuing) guarantees voice/video get served first up to a configured bandwidth limit
- Policing typically drops or re-marks traffic that exceeds a defined rate, often applied at a network edge to enforce a contracted bandwidth limit
- Shaping buffers (delays) excess traffic rather than dropping it, smoothing bursts to fit within a target rate over time - generally preferred over policing when avoiding drops matters
- Congestion management (queuing) vs congestion avoidance (e.g., WRED - Weighted Random Early Detection, which proactively drops some packets before a queue fully fills to avoid a more disruptive tail-drop event) are two different strategies
- Trust boundaries define where in the network markings from end devices/access switches are trusted vs. re-classified, since end devices could otherwise mark their own traffic as falsely high priority

Command Reference

Command	Purpose
`show mls qos interface`	Verify QoS trust state and configuration on a switch interface
`mls qos trust {cos|dscp}`	Configure a switch interface to trust incoming CoS or DSCP markings
`class-map match-any/match-all <name>`	Define traffic classification criteria for MQC-based QoS policy

Command	Purpose
`policy-map <name>`	Define actions (priority, bandwidth, police, shape) to apply to classified traffic
`service-policy {input|output} <name>`	Apply a QoS policy-map to an interface in a given direction
`priority <bandwidth-kbps>`	Configure LLQ to guarantee strict priority bandwidth for a traffic class (within policy-map class config)

Configuration Walkthrough

Basic MQC policy prioritizing voice traffic with LLQ

```
R1(config)#class-map match-all VOICE
R1(config-cmap)#match dscp ef
R1(config)#policy-map WAN_QOS
R1(config-pmap)#class VOICE
R1(config-pmap-c)#priority 256
R1(config)#interface Serial0/0/0
R1(config-if)#service-policy output WAN_QOS
```

Trust DSCP markings on a switch access port connecting to an IP phone

```
SW1(config)#interface FastEthernet0/5
SW1(config-if)#mls qos trust dscp
```

Worked Scenario

Q: A company configures QoS to prioritize VoIP traffic, but call quality issues persist even after deployment. What's a likely explanation that has nothing to do with the QoS config itself?

QoS only has an effect under congestion. If the link in question isn't actually congested, QoS marking/queuing changes nothing, and the real cause of poor call quality lies elsewhere (e.g., jitter from an unrelated source, a failing physical link, or insufficient actual bandwidth rather than a scheduling problem). Confirm there's genuine contention for bandwidth before assuming a QoS policy itself is misconfigured.

Common Mistakes / Gotchas

- Deploying QoS expecting it to help on a link that isn't actually congested
- Confusing policing (drops excess traffic) with shaping (buffers/delays excess traffic) - they solve similar problems differently
- Not establishing a trust boundary, allowing end devices to mark their own traffic as artificially high priority

Exam Tips

★ *Know DSCP (L3, 6-bit) vs CoS (L2, 3-bit) and that voice commonly maps to EF*

★ *Policing drops, shaping delays/buffers - this distinction is a frequent exam question*

★ *QoS is irrelevant without congestion - a conceptual point tested directly*